Impacts of the U.S. Inflation Reduction Act on hydrogen adoption

Quantitative summary and opportunities for Australia

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Executive summary

This report discussed the significance of the Inflation Reduction Act (IRA) and its synergy with complimentary policy to scale up clean hydrogen adoption across the U.S. The quantitative impacts of the IRA were assessed to identify competitive clean hydrogen uses and the cost gap between hydrogen producers to hydrogen users. The response of industry to the IRA is examined to understand the attitude towards investment in U.S. hydrogen projects. Opportunities for Australia to participate in reshaping the U.S. energy supply chain were explored to understand how the current Australian energy supply chain could be re-shaped through clean hydrogen.

Information was obtained from published reports, previous research and interviews with hydrogen professionals in the U.S. Interviewed stakeholders included university researchers, energy utility providers and hydrogen technology companies in Washington State.

Definitions

The IRA defines 'Clean hydrogen' as hydrogen produced with a life cycle emission intensity less than 4 kilogram carbon dioxide per kilogram hydrogen. This definition is agnostic of hydrogen source such as blue or green hydrogen.

Context

- Essentially no clean hydrogen production in the U.S. currently.
- 70% of announced hydrogen production projects are small green hydrogen (120KW – 120 MW / 50 kg – 48 T hydrogen/day) (1).
- 95% of announced hydrogen production are blue (1).

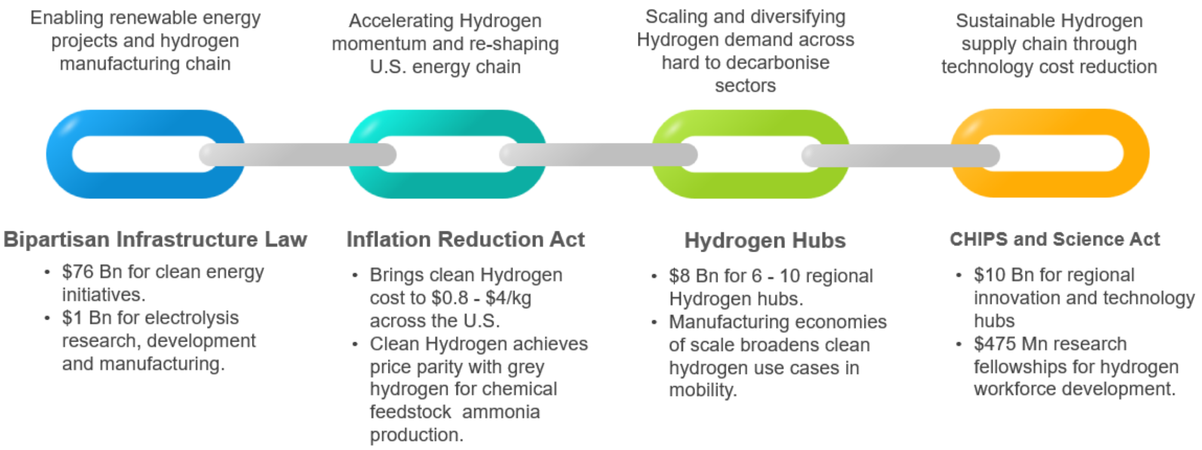
Key insights

- The IRA is one link of a broader policy chain. The full chain is enabling low hydrogen technology manufacturing and a skilled workforce ready to maximise the hydrogen hubs opportunity.
- The IRA production tax credit brings clean hydrogen to the cost threshold for substituting grey hydrogen as a chemical feedstock for ammonia and steel smelting.
- The IRA production tax credit is reshaping the U.S. energy supply chain by enabling regions without natural energy resources to produce low cost hydrogen energy.
- A cost gap exists between hydrogen production and end use sites. Costs reductions to hydrogen distribution equipment through the IRA investment tax credit is expected to enable broader hydrogen adoption in mobility and other hard to decarbonise sectors.
- Major companies are establishing hydrogen technology development projects in conjunction with Department of Energy grants. These companies are leveraging U.S. legacy hydrogen expertise to establish a skilled hydrogen workforce to maximise the hydrogen hubs opportunity.
- Expansion of electrolysis for green hydrogen production across the U.S. will require vast amounts of critical minerals that Australia has vast reserves of.
- Reshaping the U.S. energy supply chain present the opportunity for Australian hydrogen technology disruptors to accelerate commercial scale up.

Chain of policy

The IRA is one link of a greater policy chain for clean hydrogen adoption in the U.S. The pieces of policy synergise together to achieve low cost hydrogen technologies manufactured at scale, with a ready workforce to implement. These links are intended to provide the capabilities to maximise the hydrogen hubs opportunity and a sustainable hydrogen energy chain when IRA subsidies expire post 2030. The main functions of the policy links are discussed in Figure 1 and below.

Figure 1: Chain of U.S. clean hydrogen policies.



- **The Inflation Reduction Act** is providing initial momentum for clean hydrogen production and is reshaping the U.S. energy supply chain. Clean hydrogen becomes cost competitive with grey hydrogen for use adjacent to the production site.
- **The hydrogen hubs scheme** is anticipated to expand clean hydrogen adoption in hard to decarbonise sectors including mobility. Economies of hydrogen production scale and equipment manufacturing scale propagate the IRA cost savings to downstream hydrogen users to achieve a sustainable hydrogen energy chain when IRA subsidies expire in 2030.
- **The Bipartisan Infrastructure Law and CHIPS Act** focus on long term sustainability of clean hydrogen energy. Support for hydrogen technology cost reduction, domestic manufacturing scale and a trained workforce is expected to maximise the hydrogen hubs opportunity.

IRA quantitative impacts

Review of the IRA

The IRA provides tax credits for hydrogen cost reduction at the gate of the production site and across the energy distribution chain to customers. Both of these mechanisms are needed to establish clean hydrogen use across a variety of consumers ranging from chemical feedstocks to transportation fuel. Pairing these credits has the potential to disrupt the conventional U.S. energy supply chain by enabling regional hydrogen production and demand.

Hydrogen production plants can choose either a production tax credit (PTC) or investment tax credit (ITC) based on the lifecycle GHG emissions of the plant under section 45V. Both the PTC and ITC credit percentage can be increased by a multiplier of five if wage and apprenticeship conditions are met (Table 1). The IRA extends the ITC to energy storage projects including hydrogen storage vessels. The capital cost credit for energy storage projects is a 6% capital cost base rate and can reach a 30% rate if wage and apprenticeship conditions met.

Table 1: IRA PTC / ITC for hydrogen production plants with the 5x bonus applied.

Lifecycle GHG Emissions Rate (kg CO2/kg H2)	PTC $/kg of H2 Produced	ITC – Percentage of Cost
0 kg – 0.45 kg	$3	30%
0.45 kg – 1.5 kg	$1	10%
1.5 kg – 2.5 kg	$0.75	7.5%
2.5 kg – 4 kg	$0.60	6%

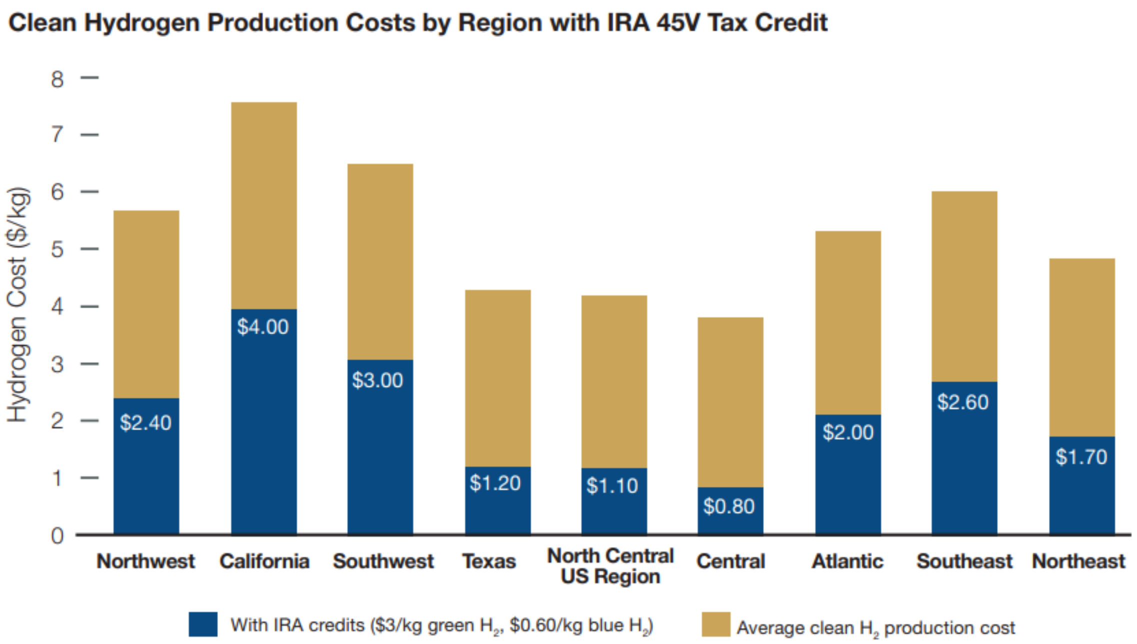
Hydrogen production plants leveraging the PTC/ITC could deliver hydrogen at a cost $0.8 - $4/kg at the gate of the facility (1). This production credit can propagate savings to co-located businesses in industrial hubs for clean hydrogen substitution as a chemical feedstock in ammonia synthesis or in green steel smelting.

Storage and transportation supply chain steps increase the delivered cost of hydrogen at regional demand sites such as hydrogen vehicle refuellers. The ITC for energy storage projects has the potential to reduce the costs incurred during hydrogen transportation process by subsidising the capital cost of bulk hydrogen storage tanks and trailers. Both subsidies synergise together and could help achieve low cost hydrogen delivery across various U.S. regions.

Regional clean hydrogen cost

The Energy Futures Initiative modelled the post-IRA cost of clean hydrogen production across various U.S. regions (1). The IRA PTC/ITC could reduce the hydrogen production cost to between $0.8/kg to $4/kg across the U.S. (Figure 2). This subsidy brings clean hydrogen to cost competitiveness with grey hydrogen ($1.3/kg) for a variety of U.S. regions. Some regions reach the threshold for clean hydrogen competitiveness in green steel smelting ($0.9/kg) and in ammonia synthesis ($0.8/kg) (1). Further cost reductions through technology development and economies of scale are needed to make clean hydrogen viable for use in refining ($0.5/kg) and chemical feedstocks like methanol ($0.2/kg).

Figure 2: Regional U.S. clean hydrogen production costs with IRA PTC.



Energy Futures Initiative modelling assumptions

- Basis of 1 MT hydrogen production per year in each region.
- Each scenario targeted the lowest cost of hydrogen and considers a mix of green and blue hydrogen production based on regional availability of renewables and carbon dioxide sequestration resources.
- Renewable mix includes solar and wind (Hydropower and other renewable sources omitted)

Reshaping the U.S. energy supply chain

The IRA green hydrogen subsidy is reshaping the energy supply chain across U.S. regions without fossil fuel resources. The example of IRA enabled energy independency in Washington State is explored below.

Washington state is the wheatbelt of the U.S. with 13% of the economy attributed to agricultural produce. The ammonia fertiliser vital to this industry is imported via train from the gulf coast and results in higher prices and supply scarcity. However, Washington state's electricity grid is powered by 60% hydro power and presents the opportunity for local, cheap green hydrogen production for mobility fuels and ammonia production.

Early clean hydrogen offtakers in Washington state are leveraging the synergy of pairing renewable power generation with green hydrogen production to minimise hydrogen production costs. Douglas County Public Utility District (PUD) is installing a pair of 5 MW electrolysers (4 Tonne hydrogen/day total) at their 880 MW hydroelectric plant (2). The electrolysers reduce utility maintenance costs by stabilising the power generated from the hydro plant while also utilising cheap, off peak solar energy across the western seaboard to produce low-cost hydrogen. The hydrogen is intended for use at a local hydrogen refuelling station for Douglas County PUD fleet vehicles. The enabling levers and impacts of the Douglas County PUD electrolyser project are summarised in Table 2. It appears that the IRA is providing initial momentum for hydrogen production, however further hydrogen costs reductions (through cost savings to existing renewable assets) is likely needed to make hydrogen production viable while local hydrogen demand is low.

Table 2: Enabling levers and impacts of the Douglas County PUD electrolyser.

Enabling levers	Clean Energy Transformation Act Signed in WA state. • All electric utilities must generate 100% of their power from renewable or zero-carbon resources by 2045.
	Authorisation of WA state PUD's to produce hydrogen • Senate Bill 5588 authorises PUD's in WA state to produce and sell hydrogen.
	Inflation Reduction Act PTC • Provides initial momentum for hydrogen cost viability, but additional cost savings through electrolyser pairing with renewable energy are needed.
Project impacts	Savings of $1.5-2 Mn/year for using hydrogen as a reserve buffer capacity for grid electricity.
	Additional savings are expected due to reduced hydroelectric turbine maintenance through use of the electrolyser to stabilise renewable energy generation.

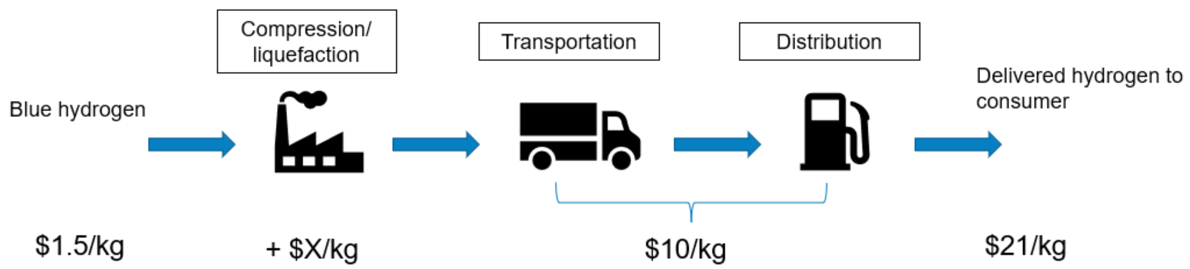
The concept of hydrogen production and renewable energy synergy could be applied in Australia to re-shape the current energy supply chain. A PTC combined with renewable energy co-location cost savings could enable low-cost regional hydrogen use in mines and farms. The synergy cost savings could come in the form of simplified logistics to mine sites where water can be trained/trucked in for local hydrogen fuel production as compared to requiring separate water and fuel transportation to site. Local hydrogen production could also achieve cost reduction synergy through acting as a energy storage buffer for renewables on a mine/farm.

Impact of ITC to lower delivered cost of hydrogen.

Hydrogen storage and transportation to consumers significantly increases the cost of hydrogen delivered beyond the range of $0.8/kg - $4/kg. The ITC for energy storage has the potential to reduce the levelized cost of hydrogen distribution through reduced capital equipment costs and increased hydrogen equipment manufacturing scales.

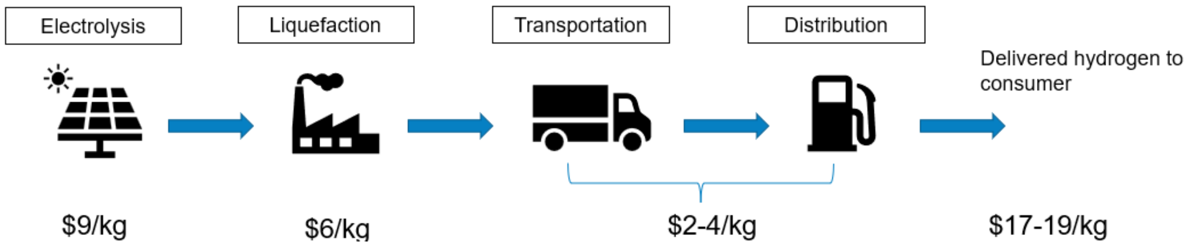
Fuel cell vehicles in California are primarily supplied by hydrogen production sites in Nevada. The hydrogen is transported via truck in either compressed gas or liquid form to refuelling stations in California. Each refuelling station has a buffer storage tank that is refilled every day or two depending on hydrogen fuel cell car demand. Figure 3 shows the hydrogen fuel supply chain from January 2023, where the delivered hydrogen price reached $21/kg due to the inflated price of natural gas feedstock and the declining value of the California low carbon fuel standard subsidy (3). Long distance distribution of hydrogen to demand sites can comprise 50% of the delivered cost of hydrogen (4). One contributing factor is the high capital cost of hydrogen storage equipment and tube trailers, as hydrogen equipment manufacturing is bespoke and has not reached sufficient economies of scale.

Figure 3: Typical supply chain costs for California hydrogen refuelling stations.



The transportation pinch point is also experienced in Australia due to our similar road-based logistics infrastructure. A feasibility study conducted by Monash University found that transportation costs from hydrogen production site to a potential hydrogen vehicle refueler location could incur a cost of $2-4/kg for hydrogen distribution within a 50km radius (Figure 4) (5). It could be expected that longer hydrogen distribution distances may incur similar transportation costs to California. The basis of this study was a demonstration scale of 5 Tonne hydrogen production per day. Achieving economies of scale at larger hydrogen production plants would increase the percentage of total cost due to transportation as hydrogen production costs reduce.

Figure 4: Expected supply chain costs for smalls scale hydrogen refuelling in Victoria.



The ITC for hydrogen storage equipment is anticipated to reduce the levelized cost of hydrogen distribution to regional demand sites through hydrogen storage equipment capital cost reductions. Both the PTC and ITC are needed to achieve low-cost hydrogen for regional consumers such as refuelling stations. A standalone IRA PTC has the potential to make clean hydrogen viable as a chemical feedstock at co-located refineries. However, an ITC could be vital to propagate the PTC cost savings to hydrogen consumers by reducing the cost of hydrogen storage equipment while equipment manufacturing is currently small scale. While the IRA ITC may reduce the cost gap between hydrogen production and consumption sites, further costs reduction to hydrogen distribution through manufacturing scale up and technology development are likely needed.

Response to IRA and supporting policy

The chain of hydrogen policies appears to be providing confidence to large hydrogen producing/consuming companies for establishing a U.S. presence. Numerous companies are presently focussed on technology development for low cost, high efficiency hydrogen storage distribution and end use (Table 3). Development of hydrogen technologies downstream of production is required for broader low-cost hydrogen adoption across mobility and import/export sectors. It may be expected that companies are currently focussed on technology development to prepare for large scale hydrogen activation as part of the hydrogen hubs scheme in 2030 onwards. Companies establishing a U.S. presence are leveraging legacy hydrogen expertise and Department of Energy grants to increase the technology readiness level of hydrogen use. Growing a U.S. presence equips these companies with a hydrogen trained U.S. workforce to realise the hydrogen hubs opportunity. However, there is still scepticism of realising large scale hydrogen hubs due to limited hydrogen skilled labour and insufficient equipment manufacturing. The general attitude amongst companies towards the opportunities of the IRA and hydrogen hubs is one of balancing 'what we have the opportunity to do' with 'what we can do'.

Table 3: Major hydrogen technology development projects in the U.S.

Company	Current focus	Notes
Shell	Demonstration of low cost, large scale liquid hydrogen storage at import/export terminals	Jointly funded through $6 Mn DOE grant
Airbus	Development of hydrogen equipment for aviation	Exploring location for U.S. hydrogen aircraft centre of excellence
Daimler (subsidiary of Mercedes)	Development of improved efficiency hydrogen fuelled trucks	Partly funded through $127 Mn DOE grant
Fortescue Future Industries	Development of renewable energy technology and opportunities to produce green hydrogen	$80 Mn partnership with U.S. National Renewable Energy Lab
Woodside	Large scale production of green hydrogen for mobility applications	Conducting front end engineering design before final investment decision

Potential opportunities for Australia

The IRA presents the opportunity for Australia to integrate into the U.S. hydrogen technology manufacturing chain, and to scale up Australian hydrogen technology disruptors.

Clean energy technology manufacturing

Reshaping the U.S. energy chain with green hydrogen requires vast amounts of critical minerals for electrolyser manufacturing. Replacing the existing 10 MT/year of grey hydrogen production in the U.S with green hydrogen would require 3.4 GW of electrolysers to be installed every year until 2050. The two main electrolysis technologies: alkaline (Alk) and proton exchange membrane (PEM) require minerals that are sparsely located across the U.S. but exist as large reserves in Australia (Table 4). The accelerating electrolyser demand in the U.S. presents the opportunity to rapidly scale Australian critical mineral extraction. A sufficient scale up of critical mineral extraction could enable higher value-added manufacturing of electrolyser components in Australia and may reduce the cost of hydrogen technologies in Australia compared to international technology imports. The IRA provides a 10% bonus on the PTC for manufactured products incurring at least 40% of their total cost in the U.S. (6). This may mean that U.S. organisations may prefer to import electrolyser parts prior to system assembly in the U.S.

Table 4: Critical mineral inputs for green hydrogen production (7), (8).

Critical mineral for electrolysis	Demand per GW of electrolysis (Ton)	Percentage of global production in the U.S. (%)	Proportion of global reserves in Australia (%)
Nickel (Alk)	2	0.55	20
Cobalt (Alk)	2.1	0.42	17
Steel (Alk)	30000	-	-
Titanium (PEM)	410	2	30

Australian hydrogen technology scale up

Regional U.S. clean hydrogen development will require new technologies for hydrogen production, storage and end use that leverage regional assets (access to sea water, natural gas, geological salt caverns etc). Australia is a leader of numerous disruptive technologies that could enable regional clean hydrogen adoption whilst utilising the U.S. as a stage for commercial scale up prior to full scale implementation in Australia (Table 5).

Table 5: Example Australian disruptor technologies with potential for scale up in the U.S.

Technology	Current commercial readiness level
Sea water electrolysis for green hydrogen production	1
Hazer clean hydrogen production from natural gas	2
Lavo solid state hydrogen storage	2
Geological hydrogen storage	1
Green steel smelting	1

Conclusion

The IRA PTC enables clean hydrogen to be cost competitive to grey hydrogen for collocated chemical industries. However, a cost gap still exists between hydrogen production sites and regional end users including hydrogen refuelling stations. Further cost reductions to hydrogen technologies and expansion of equipment manufacturing scale are required to minimise the cost gap of the supply chain. Numerous hydrogen technology companies are establishing a U.S. presence for hydrogen distribution and end use cost reduction. This U.S. presence leverages legacy hydrogen expertise to develop a hydrogen ready workforce for the hydrogen hubs opportunity in 2030 onwards. The IRA is reshaping the U.S. energy supply chain and offers the opportunity for low cost hydrogen production through co-location of green hydrogen production with renewable energy generation. The anticipated increase of green hydrogen projects will require critical minerals that Australia has large reserves of. Australia also has the opportunity to commercially scale disruptor hydrogen technologies in the U.S. to enable full scale deployment for low-cost clean hydrogen in Australia.

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